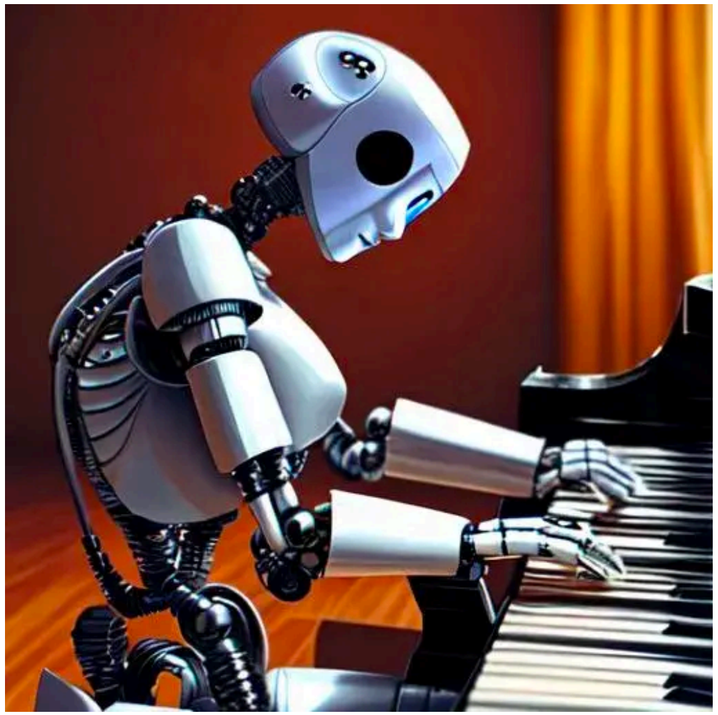
BILL E.AI.VANS

Generating jazzy + coherent melodic improvisations over a real/generated chord progression

---

Ege, Alex, Alberto - Oguz

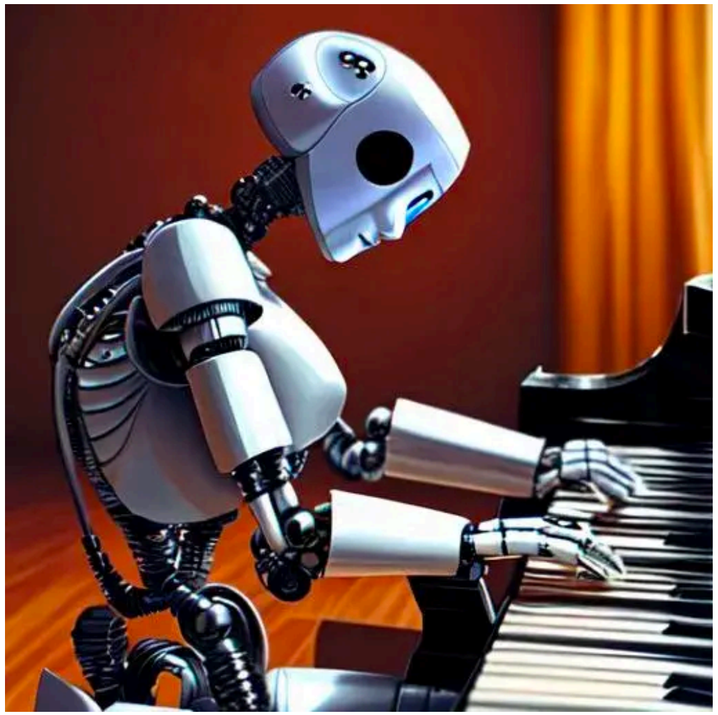


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DATASET

WEIMAR JAZZ DATABASE (WJAZZD): 450 transcribed jazz solos in midi

APPROACH

1. Treat melodies as "shapes/structures", **represent** tuples = (durations, pitches)
2. **Generate** shapes/structures
3. Harmonize the generated melodies to the generated chord progression: **Quarmonization (Rule-Based)**
4. Introduce syncopation, accomp. rhyhtm for **musical enhancement**

REPRESENTATION

music21

```
event: <music21.note.Rest 16th> M. no: 69
event: <music21.note.Note D> M. no: 69
event: <music21.note.Note B> M. no: 69
event: <music21.note.Note C#> M. no: 69
event: <music21.note.Rest 1/6ql> M. no: 69
event: <music21.note.Note B> M. no: 69
event: <music21.note.Rest 16th> M. no: 69
event: <music21.note.Note C> M. no: 69
event: <music21.note.Note B-> M. no: 69
event: <music21.note.Note B-> M. no: 70
event: <music21.note.Rest 1/6ql> M. no: 70
event: <music21.note.Rest 1/12ql> M. no: 70
event: <music21.note.Note A> M. no: 70
event: <music21.note.Rest 1/6ql> M. no: 70
event: <music21.note.Note G> M. no: 70
event: <music21.note.Rest 16th> M. no: 70
event: <music21.note.Note F> M. no: 70
event: <music21.note.Note E> M. no: 70
event: <music21.note.Rest 16th> M. no: 70
```

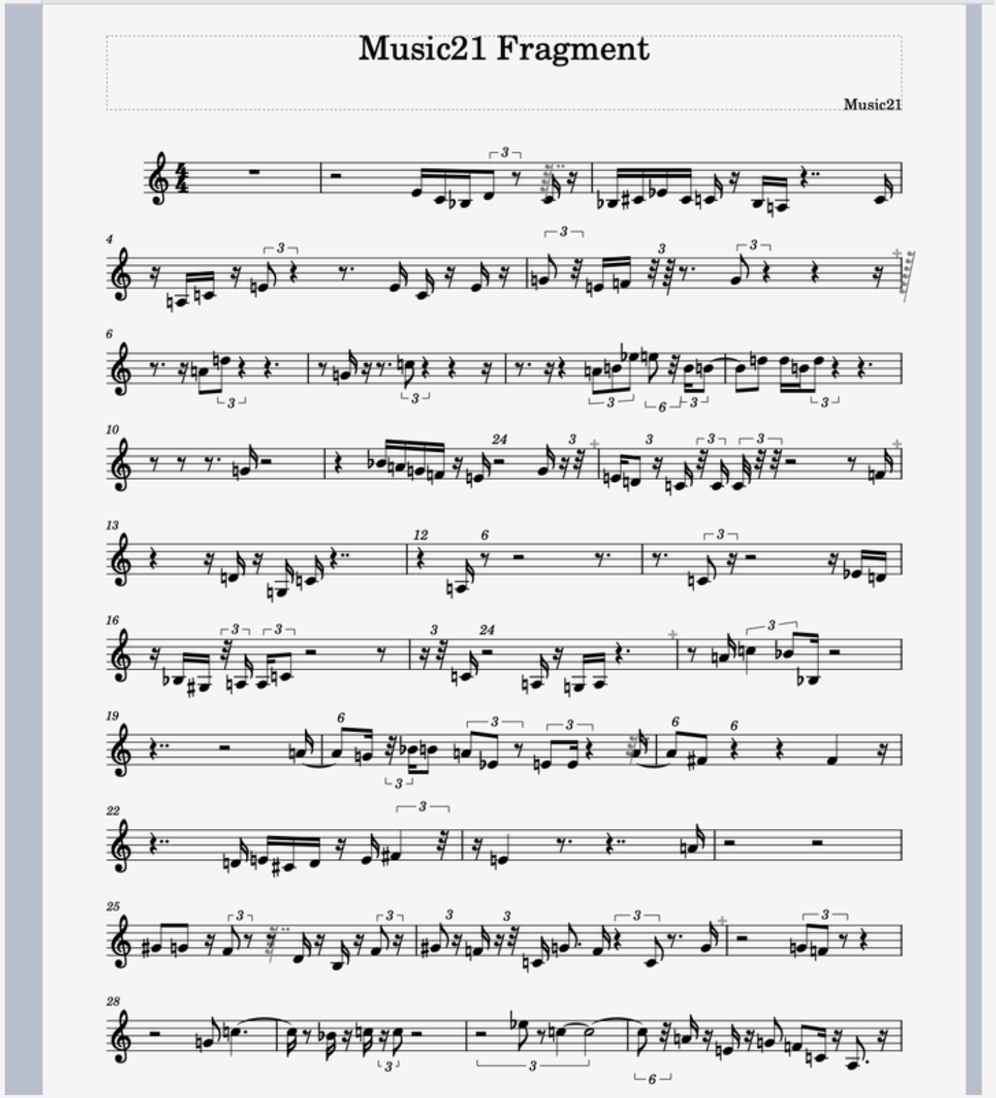
manual calculation using csv

beatdur	denom	division	duration	num	onset	period	pitch
0.3230612244897957	4	2	0.06965986394557823	4	5.131609977324263	4	60.0
0.3308616780045357	4	4	0.07401360544217687	4	5.217233560090703	4	61.0
0.3308616780045357	4	4	0.1364172335600907	4	5.308662131519275	4	62.0
0.3308616780045357	4	4	0.14222222222222222	4	5.462494331065759	4	65.0
0.32331065759637134	4	2	0.12335600907029479	4	5.780317460317461	4	64.0
0.31346938775510225	4	3	0.19156462585034015	4	5.918185941043084	4	67.0
0.31346938775510225	4	3	0.09868480725623582	4	6.134421768707483	4	70.0
0.3135147392290252	4	2	0.17560090702947845	4	6.246167800453515	4	74.0
0.3135147392290252	4	2	0.1349659863945578	4	6.439183673469388	4	73.0
0.32333333333333325	4	2	0.12625850340136055	4	6.7482993197278915	4	70.0
0.3135147392290252	4	2	0.14077097505668934	4	6.8861678004535145	4	67.0
0.3135147392290252	4	2	0.14947845804988663	4	7.031292517006802	4	68.0
0.32233560090702884	4	2	0.156734693877551	4	7.212698412698413	4	69.0
0.32233560090702884	4	2	0.13931972789115646	4	7.411519274376417	4	65.0
0.3414058956916106	4	4	0.14657596371882087	4	7.754013605442177	4	61.0
0.3310430839002265	4	4	0.1349659863945578	4	7.916553287981859	4	60.0
0.3310430839002265	4	4	0.1073922902494331	4	8.07857142857143	4	57.0
0.3407029478458057	4	4	0.10448979591836735	4	8.208163265306123	4	58.0
0.3407029478458057	4	4	0.05369614512471655	4	8.331609977324263	4	62.0
0.3407029478458057	4	4	0.10594104308390023	4	8.402721088435374	4	65.0
0.3232879818594103	4	2	0.16253968253968254	4	8.537687074829932	4	69.0

Oğuz's MTG makam code

```
[14, 10, '7560', 'C', 'C', '4', 'natural'],
[15, 0, '7560', 'C', 'C', '4', 'natural'],
[15, 1, '5040', 'B', 'B', '3', 'flat'],
[15, 2, '5040', 'G', 'G', '3', 'sharp'],
[15, 3, '3360', 'C', 'C', '4', 'natural'],
[15, 4, '2520', None, 'Rest', '', ''],
[15, 5, '3360', 'B', 'B', '3', 'natural'],
[15, 6, '3360', None, 'Rest', '', ''],
[15, 7, '3360', 'A', 'A', '3', 'natural'],
[15, 8, '5040', 'B', 'B', '3', 'flat'],
[15, 9, '1680', None, 'Rest', '', ''],
[16, 0, '6720', 'F', 'F', '3', 'sharp'],
[16, 1, '3360', 'G', 'G', '3', 'natural'],
[16, 2, '3360', 'B', 'B', '3', 'flat'],
[16, 3, '5040', 'C', 'C', '4', 'sharp'],
[16, 4, '4200', None, 'Rest', '', ''],
[16, 5, '2520', 'G', 'G', '3', 'sharp'],
[16, 6, '5040', 'C', 'C', '4', 'sharp'],
[16, 7, '5040', 'D', 'D', '4', 'natural'],
[16, 8, '5040', 'B', 'B', '3', 'flat'],
```

REPRESENTATION



Oğuz’s MTG makam code

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```

GENERATION ARCHITECTURES

for chord progression generation

1. **markov** (real book dataset)

for melodic shapes

1. **QUARMONIZATION** (**alex** is the inventor of the word)
2. generation with **perlin noise** and simple rules for **pitch and rhythm**
3. **markov** model trained on our jazz solos data
4. **VAE** trained our jazz solos data (!)

for the accompaniment (chords+drum)

1. **cellular automaton** for chord syncopation + drum

Bill Randans Raw

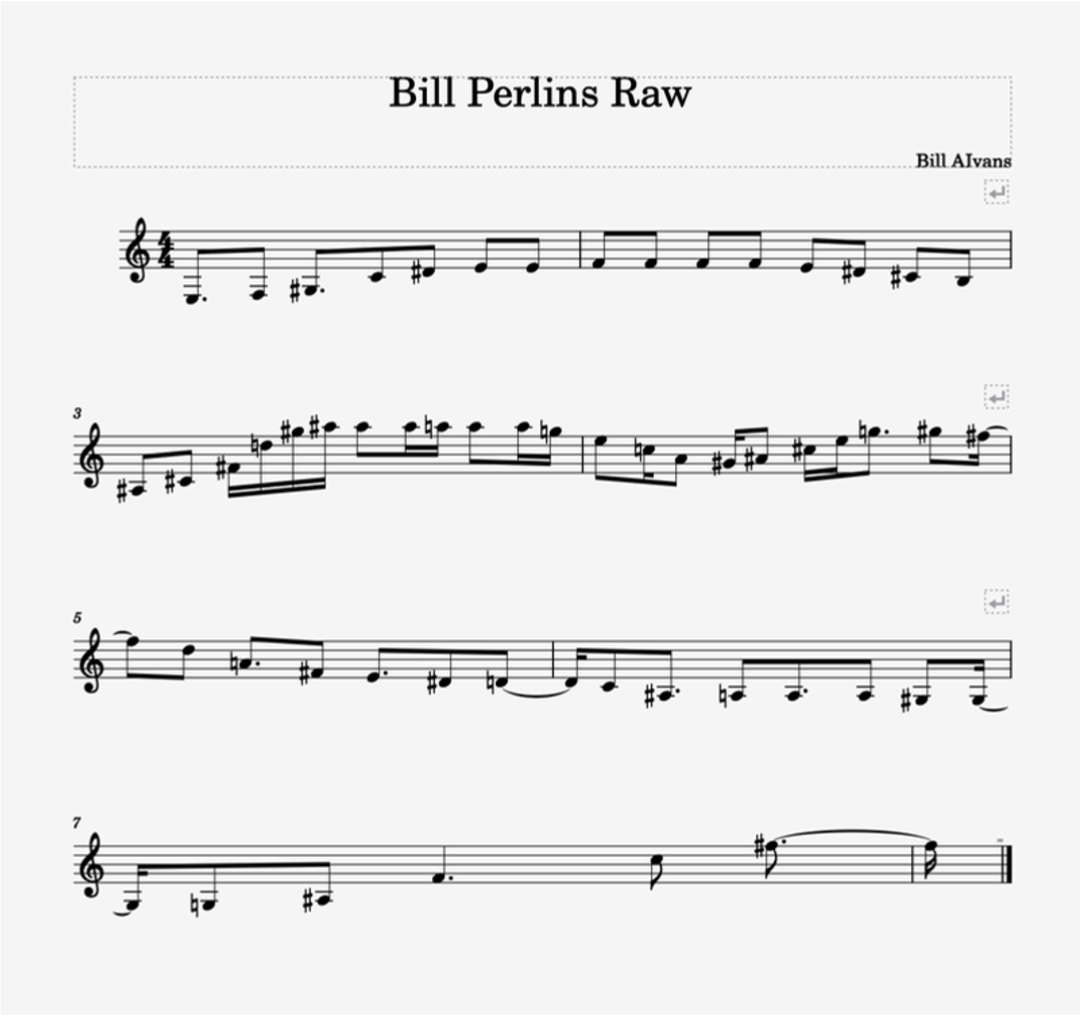
Bill AIvans



Bill Randans Quarmonized

Bill AIvans




Bill Perlins Raw
Bill AIvans


Bill Perlins Quarmonized
Bill AIvans
maj7
maj7
m7
7
maj7
7#9
m7
7
maj7

Bill Markvans Raw

Bill AIvans



Bill Markvans Quarmonized

Bill AIvans



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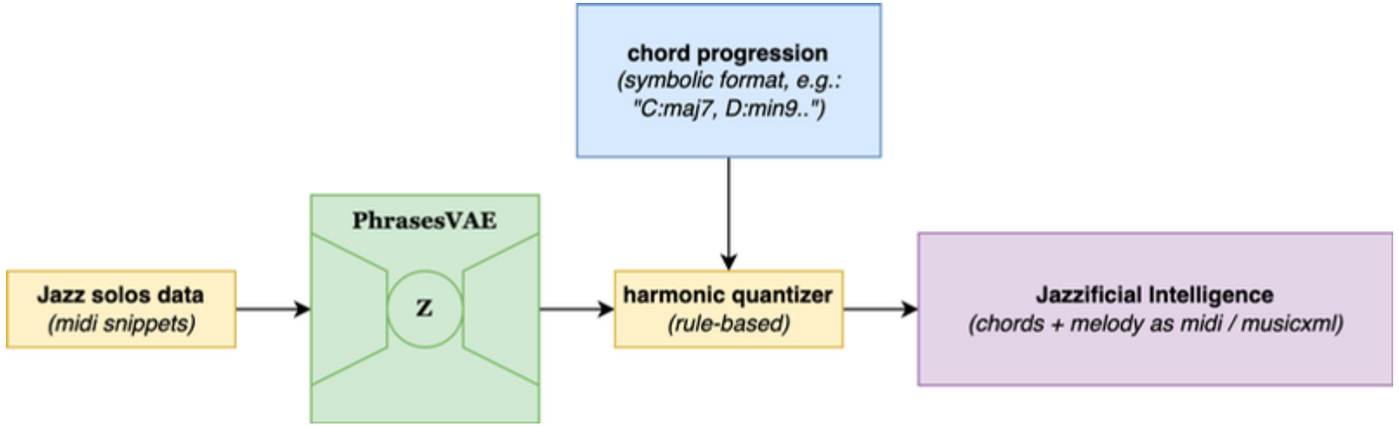
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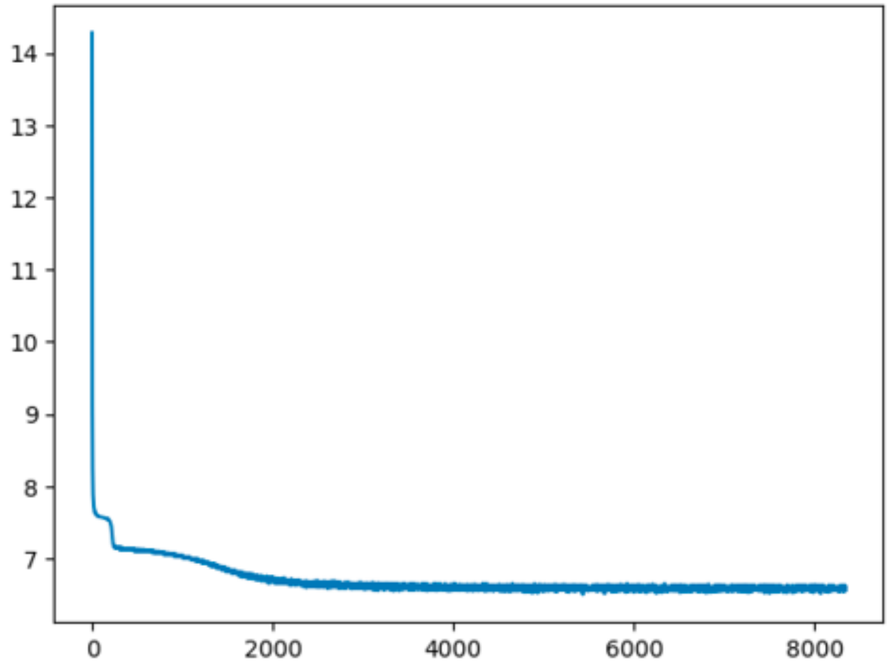
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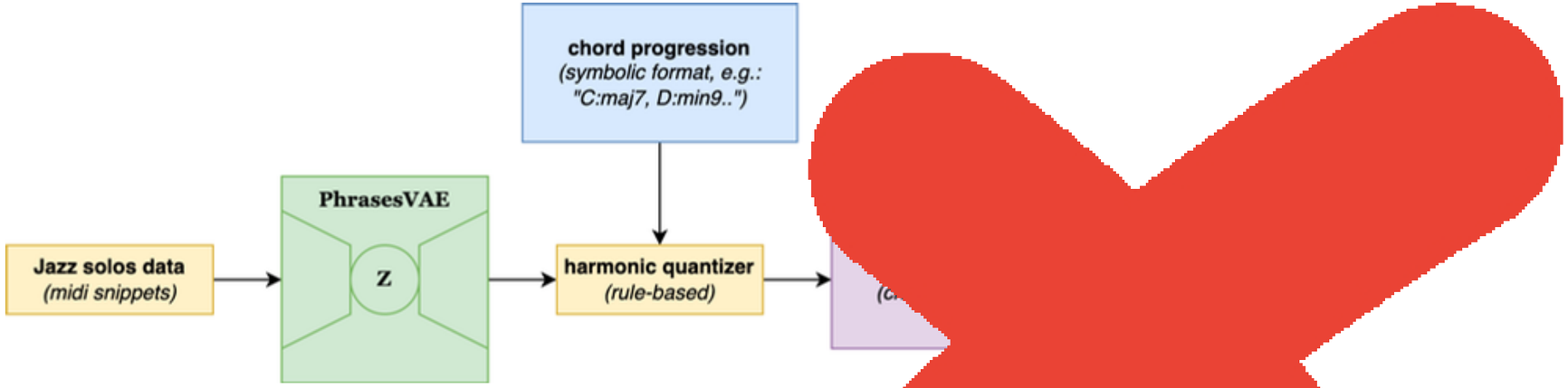
0: INITIAL IDEA



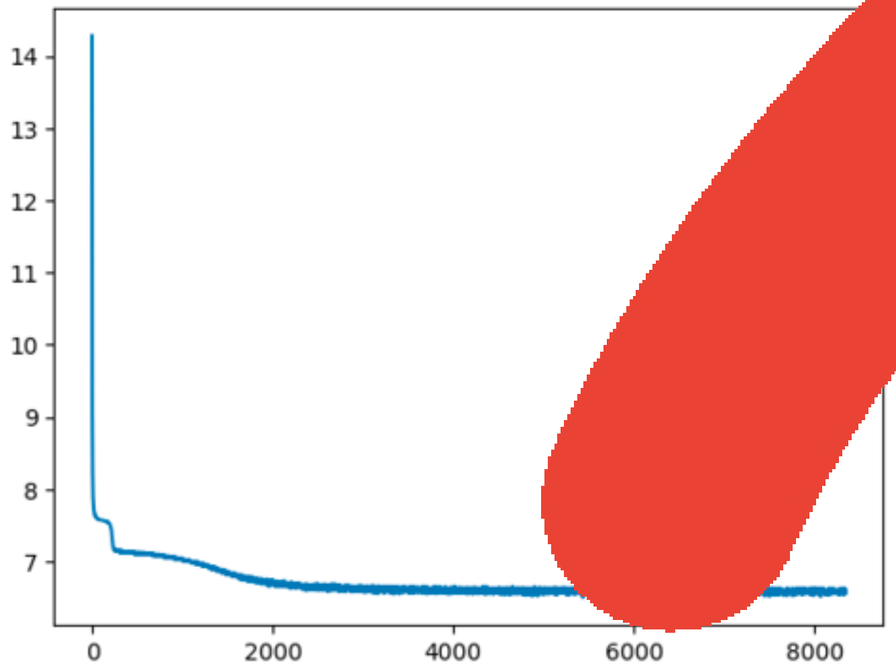
Loss Function



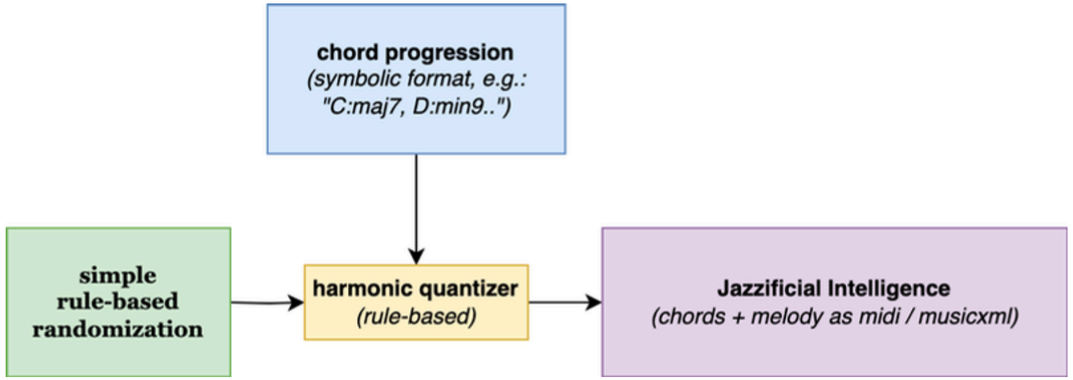
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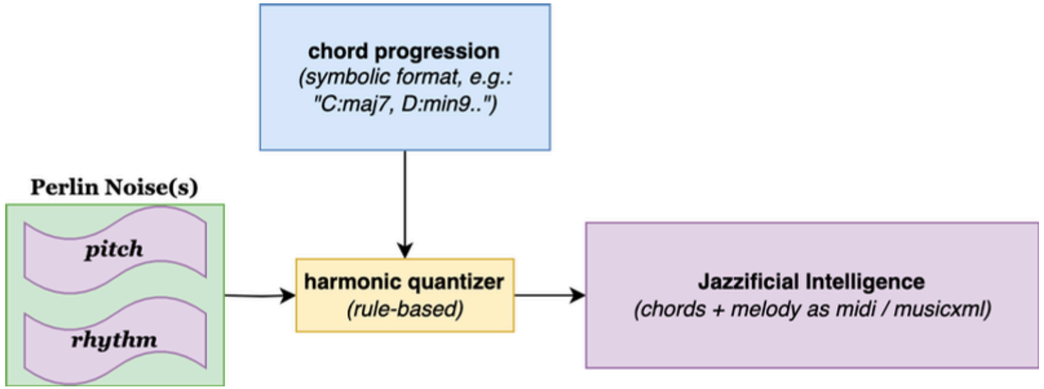
Loss Function



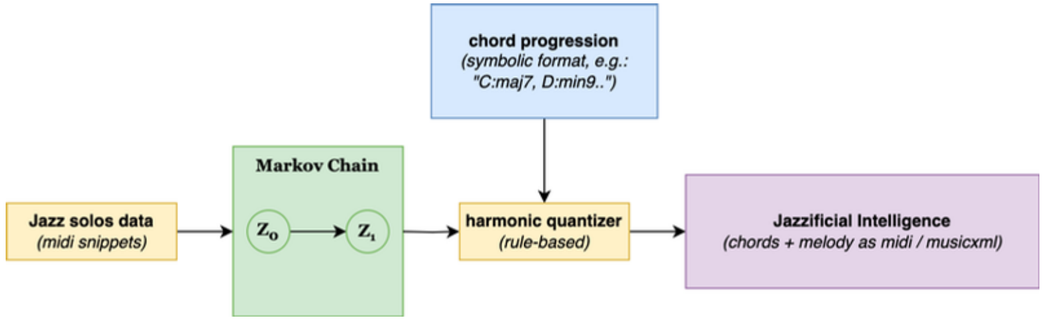
1: MELODY + SIMPLE RULE RANDOMIZATION



2: PERLIN+QUARMONIZED



3: MELODY+MARKOV+QUARMONIZED, FIXED PROGRESSION



UNHARMONIZED MELODY

HARMONIZED MELODY

MELODY+ CHORDS+RYTHM

(FIXED PROGRESSION)

SUMMARY

> 3 different work packages:

- data
- melody generation
- rhythm-syncopation generation

combined score



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> 3 different work packages:

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